NBA Shot Quality Capstone

Methodology

DAT 490 — Data Science Capstone (2026 Summer A)

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Methodology

This section describes the analytical procedures we're using to answer each of our three research questions. Following the project plan, the work is organized by research question, with each RQ getting a description of the modeling approach, the specific techniques being applied, and how we evaluate the results. As the instructor noted in the project plan check-in, the methodology section is a forecast at this stage — we're committing to a set of techniques and an evaluation strategy, and as the analysis matures we may refine specifics like model choice or feature set.

RQ1 — Which players and teams consistently outperform their expected shot value, and what distinguishes them?

RQ1 is a layered analysis rather than a single method, because the question itself has two parts: identifying overperformers, and explaining what makes them different. To answer the first part we build an expected-shot-value model; to answer the second we aggregate that model's errors and group the resulting overperformers using unsupervised techniques.

Data preparation. The base data for this question is the enriched shot table assembled by our cleaning pipeline, with one row per field goal attempt across the 2013-14 through 2025-26 tracking era. We use shot context features that the EDA confirmed are the most informative: shot distance, court coordinates, shot zone, shot clock, dribbles before the shot, touch time, defender distance, shot type, and game-state variables like period and final margin. Features that would leak the outcome (FGM, PTS, SHOT_RESULT) are excluded explicitly. The dependent variable is SHOT_MADE_FLAG, a binary indicator of whether the shot went in.

Evaluation. Because the dataset is large (roughly two million shots), a straightforward 80/20 stratified train/test split provides enough holdout data for stable performance estimates. For the final modeling pass we'll also use 5-fold cross-validation to confirm that performance is not dependent on a single split. The primary evaluation metrics are accuracy and ROC-AUC (the area under the receiver operating characteristic curve), which together measure both how often the model is right and how well it separates makes from misses across thresholds.

The techniques we're using for RQ1:

- **Logistic regression (baseline classifier).** Logistic regression predicts the probability of a binary outcome by fitting a linear combination of features through a logistic link function. It's interpretable — each feature gets a signed coefficient that tells us whether it increases or decreases the predicted probability of a made shot — and it gives us a baseline that more complex models have to clearly outperform to earn their additional complexity. Our prototype baseline on the Kaggle 2014-15 shot logs returned an accuracy of approximately 0.61 and a ROC-AUC of approximately 0.63, with SHOT_DISTANCE showing the strongest negative coefficient (farther shots are harder) and CLOSE_DEF_DIST showing the strongest positive coefficient (more defender space helps). These directions match the basketball intuition the EDA also surfaced, which gives us confidence that the model is learning real signal.
- **Gradient-boosted trees (XGBoost or LightGBM).** Logistic regression assumes feature effects are linear and additive, which limits the relationships it can capture. Shot success is not linear: the value of an extra foot of defender space depends on the shot distance, which depends on the shot clock, and so on. Gradient-boosted trees build a sequence of

small decision trees where each new tree corrects the errors of the previous ones. The ensemble captures non-linear interactions between features without us having to specify them in advance. We expect this model to outperform the logistic baseline on both accuracy and ROC-AUC, and we'll quantify the improvement so it's visible whether the added complexity is worth it.

- **Residual aggregation (per player and per team).** Once we have the trained shot-success model, we score every shot in the dataset to produce a model-expected outcome. We then compute residuals as actual makes minus model-expected makes, aggregated per player and per team. Players with consistently positive residuals are our overperformers: they make more shots than the model expected given the context of each shot they took. This is the methodological core of the EPV / shot-quality literature reviewed earlier — separating shot selection from shot-making skill.
- **K-means clustering (player archetypes).** Identifying who is overperforming is only half the question; the second half is what distinguishes them. To answer that, we cluster overperformers using playing-style features such as usage rate, shot mix by zone, position, team pace, and average shot distance. K-means partitions the players into a small number of groups by minimizing the variance within each group, and we examine the resulting clusters to give them descriptive archetype labels (for example, "primary shot creators," "low-usage corner-three specialists," "interior finishers"). Hierarchical clustering may be used as a confirmation pass to check whether the K-means partitions are robust.

RQ2 — How has shot success by court location changed across the 2013-present tracking era?

RQ2 is a descriptive longitudinal analysis rather than a predictive modeling problem. The goal is to characterize the geography of shot success in each season of the tracking era and quantify how it has changed, rather than to predict whether a specific shot will go in.

Data preparation. We use the same enriched shot table as RQ1, but here the analysis works at the season level rather than the shot level. For each of the thirteen seasons, we group shots by SHOT_ZONE_BASIC (the NBA's standard zones: restricted area, in-the-paint non-RA, mid-range, left corner 3, right corner 3, above the break 3, backcourt) and compute the field goal percentage within each zone. The EDA already produced this view of the data, which establishes both the underlying pattern and the longitudinal trend we're modeling here.

Evaluation. Because RQ2 is descriptive rather than predictive, the “evaluation” question is whether the differences we observe between seasons are statistically meaningful or could be attributed to random variation. Each season includes roughly 200,000 shots, which gives us strong statistical power to detect even modest changes, but we still want to apply formal tests.

The techniques we're using for RQ2:

- **Spatial binning by NBA zone.** Rather than treating the court as a continuous (x, y) surface, we use the league's standard categorical zones. This makes the results directly comparable to published basketball analytics, gives every interpretation a familiar vocabulary, and matches how teams and broadcasters actually discuss shot location.

- **Kernel density estimation (smoothed heat maps).** As a complement to the zone view, we use kernel density estimation to produce smooth heat maps of shot success across the entire half-court. KDE turns thousands of discrete shot points into a smooth probability surface by spreading a small kernel around each shot location. We generate one heat map per season, which then can be compared visually side-by-side to show how the geography of effective shooting has shifted.
- **Paired statistical tests across seasons.** To quantify whether zone-level FG% differences across the era are real, we use paired comparisons of zone shooting percentages across paired seasons. This captures whether the per-zone change is larger than what we would expect from natural variation alone.
- **Generalized linear model with season as a feature.** As a complementary approach, we fit a model of shot success that includes SEASON as a feature alongside shot zone and other context. The coefficient on the interaction between season and zone tells us how the effect of location on shot success has shifted across the tracking era. This provides a single, condensed numeric summary of the longitudinal trend that supports the descriptive heat-map visualizations.

RQ3 — Does shot-quality-adjusted player ranking differ noticeably from traditional efficiency metrics (FG%, eFG%, TS%)?

RQ3 is a comparison of two rankings — one based on traditional efficiency metrics, one based on a shot-quality-adjusted metric we derive from the RQ1 model — and a test of whether they tell meaningfully different stories about which players are most efficient.

Data preparation. From the RQ1 model, we construct a shot-quality-adjusted player metric. The most natural construction is expected FG% above league average: for each player, we sum their model-expected makes and compare to their actual makes, normalized by attempts. We then compute the three traditional efficiency metrics on the same player population:

- **Field Goal Percentage (FG%).** Made shots divided by attempted shots; the most basic measure.
- **Effective Field Goal Percentage (eFG%).** FG% adjusted to give threes proper credit; (FGM + 0.5 × 3PM) / FGA.
- **True Shooting Percentage (TS%).** The most complete traditional measure, includes free throws and properly weights threes: PTS / (2 × (FGA + 0.44 × FTA)).

These three are well-established in the public basketball-analytics vocabulary and represent the metrics we're testing our shot-quality-adjusted measure against.

Evaluation. RQ3 is a ranking comparison, so evaluation is about whether the orderings are similar or different.

The techniques we're using for RQ3:

- **Spearman rank correlation.** Spearman's rho measures how similar two rankings are on a scale from -1 to +1. A value near +1 means the two rankings are nearly identical, a value near 0 means they're effectively unrelated, and a negative value would mean the rankings are inverted. Computing Spearman's rho between our adjusted metric and each of FG%, eFG%, and TS% gives us a direct quantitative answer to the question of whether the rankings differ.

- **Slope chart / rank-shift visualization.** A scalar correlation tells us how similar the rankings are on average, but it doesn't tell us which players move the most. The slope chart plots two parallel rank axes — players ranked by a traditional metric on the left, players ranked by the adjusted metric on the right — with lines connecting each player's two ranks. Players with the largest slopes are the ones whose evaluation changes most under the shot-quality lens, and those are the cases worth interpreting.
- **Subgroup analysis by player role.** Beyond the headline correlation and visualization, we look at whether certain player types — for example, primary creators, low-usage role players, interior bigs — are systematically re-ranked. This is what gives the comparison interpretive weight rather than just a numeric summary.

Computational methods summary

The complete list of statistical and computational techniques used across all three research questions:

- Logistic regression
- Gradient-boosted trees (XGBoost or LightGBM)
- Residual analysis (player and team aggregations of model error)
- K-means clustering and hierarchical clustering (player archetype identification)
- Kernel density estimation (smoothed shot success heat maps)
- Paired statistical tests (longitudinal comparison of zone-level shooting percentages)
- Generalized linear modeling with categorical interactions (season-by-zone effect)
- Spearman rank correlation (cross-ranking comparison)
- Slope-chart and rank-shift visualizations

All source code implementing these techniques — including model training pipelines, residual aggregation, clustering, heat-map generation, and the rank comparison utilities — will be included in the Appendix of the Final Report rather than in the body, per the assignment specification.